

**Problem 1: Vectors & Planes Review****(a)**

Provide a formula for computing the angle between two nonzero  $d$ -dimensional vectors  $\bar{x}^{(1)}$  and  $\bar{x}^{(2)}$ . Now suppose  $\bar{x}^{(1)} = [-\frac{\sqrt{3}}{4}, \frac{1}{4}]^T$  and  $\bar{x}^{(2)} = [\frac{\sqrt{3}}{2}, \frac{1}{2}]^T$ . What is the magnitude of each vector, and what is the angle between them in degrees?

$$\bar{x}^{(1)} \cdot \bar{x}^{(2)} = \|\bar{x}^{(1)}\| \|\bar{x}^{(2)}\| \cos \theta$$

$$\cos \theta = \frac{\bar{x}^{(1)} \cdot \bar{x}^{(2)}}{\|\bar{x}^{(1)}\| \|\bar{x}^{(2)}\|}$$

$$\theta = \arccos \left( \frac{\bar{x}^{(1)} \cdot \bar{x}^{(2)}}{\|\bar{x}^{(1)}\| \|\bar{x}^{(2)}\|} \right)$$

$$\|\bar{x}^{(1)}\| = \sqrt{\left(-\frac{\sqrt{3}}{4}\right)^2 + \frac{1}{4^2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$$\|\bar{x}^{(2)}\| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2^2}} = 1$$

$$\theta = \arccos \left( \frac{-\frac{1}{4}}{\frac{1}{2}} \right) = \arccos \left( \frac{-1}{2} \right) = 120^\circ$$